

LIE SYMMETRY PRESERVATION AND SHOCK-CAPTURING METHODS*

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Abstract. It is known that some traditional shock-capturing schemes, including Godunov- or Roe-type schemes, generate downstream oscillatory wave patterns that cannot be effectively dampened by the dissipation methods they provide. The theme of this paper is to understand the formation and behavior at these upstream/downstream patterns from the point view of the symmetry inherent in the PDE. This paper investigates symmetry breaking and spatial oscillations in finite difference solutions for the inviscid Burgers and one-dimensional Navier–Stokes shock equations using the Lax–Wendroff, TVD, and nonoscillatory, nonfree parameter dissipative schemes. Symmetry breaking is considered as a candidate for the inherent cause of the nonphysical oscillations in the vicinity of the shock.

Key words. difference scheme, symmetry, shock-capturing method

AMS subject classifications. 22E70, 39A11, 65M06

DOI. 10.1137/050648201

1. Introduction. The past four decades have seen tremendous progress in the development of numerical methods for solving the equations governing the unsteady flow of an inviscid, adiabatic ideal gas, that is, the Euler equations [1, 2, 3, 4, 5]. The goal of shock-capturing methods is to capture discontinuities in solutions to these equations automatically, without explicitly tracking them. Discontinuities must then be smeared over one or more grid cells. Success requires that the method implicitly incorporates the correct jump conditions, reduces smearing to a minimum, and does not introduce nonphysical oscillations near the discontinuities. High-resolution finite difference/volume methods based on Riemann solutions often perform well and are simple to implement. There are several circumstances where even first order schemes (such as the Godunov method and the Roe scheme) may produce inferior results [6]. One known example is the computation of slow-moving shocks. Several references have reported the difficulty of computing slow-moving shocks, where first order Godunov- or Roe-type methods produce spurious long wave oscillations behind the shock [7, 8, 9]. These eventually ruin the downstream solution. A full understanding of the nature of this kind of numerical phenomenon is still lacking. Regardless of definition, spurious oscillations and overshoots are the most common symptoms of numerical instability. In one natural interpretation, these numerical phenomena are due to nonlinear stability. Linear stability theory has a strong history, dating back to the late 1940s, but the study of nonlinear stability is far newer. Starting with papers by Boris and Book [10] and van Leer [11], nonlinear stability theory has developed over roughly the past thirty years. Perhaps the best known nonlinear stability condition is the total variation diminishing (TVD) condition, suggested by Harten in 1983 [2]. A review of this can be found in [12]. This paper addresses the interesting problem of nonlinear instability from a new point of view. Before moving on, it should be noted that the Euler equations and most scalar conservation laws are highly nonlinear.

*Received by the editors December 22, 2005; accepted for publication (in revised form) August 8, 2007; published electronically January 23, 2008. This research was supported by the National Natural Science Foundation of China under grants 10272018 and 10572083.

<http://www.siam.org/journals/sinum/46-1/64820.html>

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Numerical approximations for these should contain at least as much nonlinearity as the governing equations. Most modern numerical methods add their own nonlinearity, making them highly nonlinear even for linear governing equations. Unfortunately, in general, there are still many unresolved issues in nonlinear stability theory. There are two primary drawbacks in the existing theoretical development: heuristics and one-dimensional analysis. Indeed, both of these reflect the difficulties presented by high nonlinearity. It is hoped that Lie symmetry analysis can be used to throw some light on this subject.

Symmetry has always played an important role in mechanics, from the fundamental formulation of basic principles to concrete application [13, 14, 15]. Applications of symmetry groups to numerical analysis are given extensive treatment by Shokin [16] and Dorodnitsyn [17]. For gas dynamics, the noninvariance relative to Galileo transformation of a difference scheme which approximates the equations results in nonphysical fluctuations, which was remarked in the 1960s (Harlow 1964).¹ In 1970 Yanenko and Shokin [18] developed a method of differential approximations for the study of the group properties of difference schemes for hyperbolic systems of equations. They used the first differential approximation to perform a group analysis. The study of symmetry in difference equations is a much more recent phenomenon than for differential equations. Early work in this direction performed by Maeda [19] concentrates on transformations acting only on dependent variables. A more recent series of papers was devoted to the Lie point symmetries of differential difference equations on fixed regular lattices [20]. A different approach was developed mainly for linear or linearizable difference equations and involved transformations acting on more than one point of the lattice [21, 22]. The symmetries considered in this approach are really generalized; however, they reduce to point symmetries at the continuous limit. A more general class of generalized symmetries has also been investigated for difference equations, and differential-difference equations on fixed regular lattices [23, 24]. A further approach to investigating symmetries in discrete equations was originally suggested by Dorodnitsyn and collaborators [25, 26]. The main aim of this series of papers is to make differential equations discrete while preserving their Lie point symmetries. Symmetries of ordinary and partial difference schemes on lattices that are a priori given, but are allowed to transform under point transformations, were studied in [27, 28, 29]. In a series of lectures, presented at the CIMPA Winter School on Discrete Integrability Systems in February 2003, Winternitz [30] gave a useful review of the application of Lie point symmetries, and their generalizations, to the study of difference equations. The overall theme could be called “continuous symmetries of discrete equations.”

Three important questions on numerical analysis, especially important for shock-capturing methods are discussed from the point of view of group theory below. They are

- (1) the group classification of the Lax–Wendroff scheme,
- (2) symmetry and Harten’s entropy enforcement condition,
- (3) the role of symmetry in various aspects of a proper manipulation of the coefficients of the third order derivative ν_3 in the nonoscillatory, nonfree parameter dissipative (NND) scheme.

The structure of this paper is as follows. First, the theory of symmetries of differential equations is very briefly discussed, by way of a theoretical introduction in section 2. The group classification of the nonlinear wave equations is discussed in section 3. The following three sections are devoted to a complete development of

¹Thanks to the referee’s information.

the Lie symmetry analysis method proposed here and its application to three special cases of interest. The final section contains concluding remarks.

2. Mathematical preliminaries. All of the problems addressed here can be described by a general system of nonlinear differential equations of the n th order

$$(2.1) \quad \Delta_\nu(x, u^{(n)}) = 0,$$

where $\nu = 1, \dots, l$, and $x = (x^1, \dots, x^p) \in X$ are independent variables, $u = (u^1, \dots, u^q) \in U$ are dependent variables, and $\Delta_\nu(x, u^{(n)}) = (\Delta_1(x, u^{(n)}), \dots, \Delta_l(x, u^{(n)}))$ is a smoothing function that depends on x, u , and derivatives of u up to order n with respect to $x^1, \dots, x^p$. If we define a jet space $X \times U^{(n)}$ as a space whose coordinates are independent variables, dependent variables, and derivatives of dependent variables up to order n , then Δ is a smoothing mapping

$$(2.2) \quad \Delta : X \times U^{(n)} \rightarrow R^l.$$

Before studying the symmetries of difference schemes, let us very briefly review the theory of symmetries for differential equations. For all details, proofs, and further information we refer to the many excellent books on the subject, e.g., [13, 14, 15]. Here we follow the style of [13], but the Lie symmetry description is made concise by emphasizing the significant points and results. In order to provide the reader with a relatively quick and painless introduction to Lie symmetry theory, some important concepts must be introduced.

The main tool used in Lie group theory and working with transformation groups is “infinitesimal transformation.” In order to present this, we need first to develop the concept of a vector field on a manifold. We begin with a discussion of tangent vectors. Suppose C is a smooth curve on a manifold M , parameterized by

$$(2.3) \quad \phi : I \rightarrow M,$$

where I is a subinterval of R . In local coordinates $x = (x^1, \dots, x^p)$, C is given by p smoothing functions

$$(2.4) \quad \phi(\varepsilon) = (\phi^1(\varepsilon), \dots, \phi^p(\varepsilon))$$

of the real variable ε . At each point $x = \phi(\varepsilon)$ of C the curve has a tangent vector, namely, the derivative

$$(2.5) \quad \dot{\phi} = \frac{d\phi}{d\varepsilon} = \left(\frac{d\phi^1}{d\varepsilon}, \dots, \frac{d\phi^p}{d\varepsilon} \right).$$

In order to distinguish between tangent vectors and local coordinate expressions for a point on the manifold, we adopt the notation

$$(2.6) \quad V = \frac{d\phi}{d\varepsilon} = \frac{d\phi^1}{d\varepsilon} \cdot \frac{\partial}{\partial x^1} + \dots + \frac{d\phi^p}{d\varepsilon} \cdot \frac{\partial}{\partial x^p}$$

for the vector tangential to C at $x = \phi(\varepsilon)$. The collection of all tangent vectors to all possible curves passing through a given point x in M is called the tangent space to M at x and is denoted by TM . A vector field V on M assigns a tangent vector $V \in TM$ to each point $x \in M$, with V varying smoothly from point to point. In local coordinates, a vector field has the form

$$(2.7) \quad V = \xi^1(x) \cdot \frac{\partial}{\partial x^1} + \dots + \xi^p(x) \cdot \frac{\partial}{\partial x^p},$$

where each $\xi^i(x)$ is a smoothing function of x .

If V is a vector field, we denote the parameterized maximal integral curve passing through x in M by $\Psi(\varepsilon, x)$ and call Ψ the flow generated by V . Thus for each x in M , and ε in some interval I_x containing 0, $\Psi(\varepsilon, x)$ is a point on the integral curve passing through x in M . The flow of a vector field has the basic properties

$$(2.8) \quad \Psi(\delta, \Psi(\varepsilon, x)) = \Psi(\delta + \varepsilon, x)$$

for all $\delta, \varepsilon \in R$ such that both sides of equation are defined:

$$(2.9) \quad \Psi(0, x) = x,$$

and

$$(2.10) \quad \frac{d}{d\varepsilon} \Psi(\varepsilon, x) = V$$

for all ε where defined. We see that the flow generated by a vector field is the same as a local group action of the Lie group on the manifold M , often called a “one-parameter group of transformations.” The vector field V is called the infinitesimal generator of the action since, by Taylor’s theorem, in local coordinates

$$(2.11) \quad \Psi(\varepsilon, x) = x + \varepsilon \xi(x) + O(\varepsilon^2),$$

where $\xi = (\xi^1, \dots, \xi^p)$ are the coefficients of V . The orbits of the one-parameter group action are the maximal integral curves of the vector field V .

DEFINITION 1. A symmetry group of (2.1) is a one-parameter group of transformations G , acting on $X \times U$, such that if $u = f(x)$ is an arbitrary solution of (2.1) and $g_\varepsilon \in G$, then $g_\varepsilon \cdot f(x)$ is also a solution of (2.1).

The infinitesimal generator of a symmetry group is called an infinitesimal symmetry. Infinitesimal generators are used to formulate the conditions for a group G to make it a symmetry group. Working with infinitesimal generators is simple. First we define a prolongation of a vector field. The symmetry group of a system of differential equations is the largest local group of transformations acting on the independent and dependent variables of the system such that it can transform one system solution to another. The main goal of Lie symmetry theory is to determine a useful, systematic, computational method that explicitly determines the symmetry group of any given system of differential equations. The search for the symmetry algebra L of a system of differential equations is best formulated in terms of vector fields acting on the space $X \times U$ of independent and dependent variables. The vector field tells us how the variables x, u transform. We also need to know how the derivatives, that is, $u_x, u_{xx}, \dots$, transform. This is given by the prolongation of the vector field V . Combining these, we have [13, p. 110, Theorem 2.36] the following.

THEOREM 1. Let

$$V = \sum_{i=1}^p \xi^i(x, u) \partial_{x^i} + \sum_{a=1}^q \eta_a(x, u) \partial_{u^a}$$

be a vector defined on an open subset $M \subset X \times U$. The n th prolongation of the original vector field is the vector field:

$$pr^{(n)}V = V + \sum_{a=1}^q \sum_J \eta_a^J(x, u^{(n)}) \frac{\partial}{\partial u_J^a}$$

defined on the corresponding jet space $M^{(n)} \subset X \times U^{(n)}$. The second summation here is over all (unordered) multiindices $J = (j_1, j_2, \dots, j_k)$, with $1 \leq j_k \leq p$, $1 \leq k \leq n$. The coefficient functions ϕ_a^J of $pr^{(n)}V$ are given by the following formula:

$$\eta_a^J(x, u^{(n)}) = D_J \left(\eta_a - \sum_{i=1}^p \xi^i u_i^a \right) + \sum_{i=1}^p \xi^i u_{J,i}^a,$$

where $u_i^a = \frac{\partial u^a}{\partial x^i}$, $u_{J,i}^a = \frac{\partial u^a}{\partial x^i}$, and D_J is the total derivative of η with respect to x^j .

In the following analysis, we deal only with one-dimensional scalar differential equations that are assumed to be differentiable up to the necessary order.

Consider the special case where $p = 2, q = 1$ in the prolongation formula, so that we are looking at a partial differential equation involving the function $u = f(x, t)$. A general vector field on $X \times U \cong R^2 \times R$ then takes the form [13, p. 114]:

$$(2.12) \quad V = \xi(x, t, u) \frac{\partial}{\partial x} + \tau(x, t, u) \frac{\partial}{\partial t} + \eta(x, t, u) \frac{\partial}{\partial u}.$$

The first prolongation of V is the vector field:

$$(2.13) \quad pr^{(1)}V = V + [\eta_x] \frac{\partial}{\partial u_x} + [\eta_t] \frac{\partial}{\partial u_t},$$

where

$$[\eta_x] = \eta_x + (\eta_u - \xi_x)u_x - \tau_x u_t - \xi_u u_x^2 - \tau_u u_x u_t$$

and

$$[\eta_t] = \eta_t + (\eta_u - \tau_t)u_t - \xi_t u_x - \tau_u u_t^2 - \xi_u u_x u_t.$$

The subscripts on η, ξ, τ denote partial derivatives. Similarly,

$$(2.14) \quad pr^{(2)}V = pr^{(1)}V + [\eta_{xx}] \frac{\partial}{\partial u_{xx}} + [\eta_{xt}] \frac{\partial}{\partial u_{xt}} + [\eta_{tt}] \frac{\partial}{\partial u_{tt}},$$

where

$$\begin{aligned} [\eta_{xx}] &= \eta_{xx} + (2\eta_{xu} - \xi_{xx})u_x - \tau_{xx}u_t + (\eta_{uu} - 2\xi_{xu})u_x^2 - 2\tau_{xu}u_x u_t - \xi_{uu}u_x^3 \\ &\quad - \tau_{uu}u_x^2 u_t + (\eta_u - 2\xi_x)u_{xx} - 2\tau_x u_{xt} - 3\xi_u u_{xx}u_x - \tau_u u_{xx}u_t - 2\tau_u u_{xt}u_t, \\ [\eta_{xt}] &= \eta_{xt} + (\eta_{xu} - \tau_{tx})u_t + (\eta_{tu} - \xi_{tx})u_x - \tau_{xu}u_t^2 + (\eta_{uu} - \xi_{xu} - \tau_{ut})u_x u_t \\ &\quad - \xi_{tu}u_x^2 - \tau_{uu}u_x u_t^2 - \xi_{uu}u_t u_x^2 - \tau_x u_{tt} + (\eta_u - \xi_x - \tau_t)u_{xt} - \xi_t u_{xx} - 2\tau_u u_t u_{xt} \\ &\quad - 2\xi_u u_x u_{xt} - \tau_u u_x u_{tt} - \xi_u u_t u_{xx}, \\ [\eta_{tt}] &= \eta_{tt} + (2\eta_{tu} - \tau_{tt})u_t - \xi_{tt}u_x + (\eta_{uu} - 2\tau_{tu})u_t^2 - 2\xi_{tu}u_x u_t - \tau_{uu}u_x^3 \\ &\quad - \tau_{uu}u_t^2 u_x + (\eta_u - 2\tau_t)u_{tt} - 2\xi_t u_{xt} - 3\tau_u u_{tt}u_t - \xi_u u_{tt}u_x - 2\xi_u u_{xt}u_t. \end{aligned}$$

From here on the analysis of difference equations concerns only modified equations from the third prolongation of the vector field. From work in computational fluid dynamics, we know that the right-hand side of the modified equation is written entirely in terms of x derivatives. So investigation can be limited to the terms of the spatial derivatives in the following analysis. The coefficients of the various monomials in the third order partial derivatives of u are given in the following:

$$(2.15) \quad pr^{(3)}V = pr^{(2)}V + [\eta_{xxx}] \frac{\partial}{\partial u_{xxx}} + [\eta_{xxt}] \frac{\partial}{\partial u_{xxt}} + [\eta_{xtt}] \frac{\partial}{\partial u_{xtt}} + [\eta_{ttt}] \frac{\partial}{\partial u_{ttt}},$$

where

$$\begin{aligned} [\eta_{xxx}] = & \eta_{xxx} + (3\eta_{xxu} - \xi_{xxx})u_x - \tau_{xxx}u_t + 3(\eta_{xuu} - \xi_{xxu})u_x^2 - 3\tau_{xxu}u_xu_t \\ & + (\eta_{uuu} - 3\xi_{xuu})(u_x)^3 + 3(\eta_{xu} - \xi_{xx})u_{xx} - 3\tau_{xx}u_{xt} - 3\tau_{xuu}(u_x)^2u_t \\ & + 3(\eta_{uu} - 3\xi_{xu})u_xu_{xx} - 3\tau_{xu}u_tu_{xx} - 6\tau_{xu}u_{xt}u_x - 3\tau_xu_{xxt} + (\eta_u - 3\xi_x)u_{xxx} \\ & - \xi_{xxx}(u_x)^4 - 6\xi_{uu}(u_x)^2u_{xx} - 3\tau_{uu}(u_x)^2u_{xt} - \tau_{uuu}(u_x)^3u_t - 3\xi_u(u_{xx})^2 \\ & - 3\tau_uu_{xxt}u_x - 3\tau_uu_{xt}u_{xx} - 3\tau_{uu}u_{xx}u_xu_t - 4\xi_uu_{xxx}u_x - \tau_uu_{xxx}u_t. \end{aligned}$$

Suppose we are given an n th order system of differential equations or, equivalently, a subvariety of the jet space $M^{(n)} \subset X \times U^{(n)}$. A symmetry group of this system is a local transformation G acting on $M \subset X \times U$ which transforms solutions of the system to other solutions. We can reduce the important infinitesimals condition for a group G to be a symmetry group of a given system of differential equations. The following theorem [13, p. 104, Theorem 2.31] provides the infinitesimal conditions for a group G to be a symmetry group.

THEOREM 2. *Suppose*

$$\Delta_\nu(x, u^{(n)}) = 0, \quad \nu = 1, 2, \dots, l,$$

is a system of differential equations of maximal rank defined over $M \subset X \times U$. If G is a local group of transformations acting on M , and

$$pr^{(n)}V \circ [\Delta_\nu(x, u^{(n)})] = 0, \quad \nu = 1, 2, \dots, l,$$

whenever

$$\Delta_\nu(x, u^{(n)}) = 0,$$

for every infinitesimal generator V of G , then G is a symmetry group of the system.

In the following sections, this theorem is used to deduce explicitly different infinitesimal conditions for specific problems. It must be remembered, however, that, in all cases, though only the scalar differential problem is being discussed, Δ_ν is still used to denote different differential equations.

3. The group classification of the nonlinear wave equations. The main prototype equation here is the inviscid Burgers equation. Its mathematical properties are very clear, and its group classification can be obtained from previous work. In 1981 Ames, Lohner, and Adams [34] (also see [14, p. 194]) considered the nonlinear wave equation:

$$(3.1) \quad u_{tt} = (c^2(u)u_x)_x,$$

where $c(u) \neq \text{const}$. Viewing this in terms of infinitesimal generators reveals that the group classification of (3.1) is

a. $c(u)$ arbitrary:

$$(3.2) \quad V_1 = \partial_x,$$

$$(3.3) \quad V_2 = \partial_t,$$

$$(3.4) \quad V_3 = x\partial_x + t\partial_t.$$

b. $c(u) = A(u + B)^C$, where A, B, C are arbitrary constants:

$$(3.5) \quad V_1, V_2, V_3,$$

$$(3.6) \quad V_4 = Cx\partial_x + (u + B)\partial_u.$$

c. $c(u) = A(u + B)^{-2}$, where A, B are arbitrary constants:

$$(3.7) \quad V_1, V_2, V_3, V_4 (C = -2),$$

$$(3.8) \quad V_5 = t^2\partial_t + t(u + B)\partial_u.$$

d. $c(u) = A(u + B)^{-\frac{2}{3}}$, where A, B are arbitrary constants:

$$(3.9) \quad V_1, V_2, V_3, V_4 \left(C = -\frac{2}{3} \right),$$

$$(3.10) \quad V_5 = x^2\partial_x - 3x(u + B)\partial_u.$$

For the inviscid Burgers equation:

$$(3.11) \quad u_t + uu_x = 0.$$

This equation can be rewritten as:

$$(3.12) \quad u_{tt} = (u^2 u_x)_x.$$

The corresponding wave velocity is then:

$$(3.13) \quad c(u) = u,$$

with the corresponding classification coefficients:

$$(3.14) \quad A = 1,$$

$$(3.15) \quad B = 0,$$

$$(3.16) \quad C = 1.$$

This indicates that there are four spans in the infinitesimal symmetry group of non-linear wave equation vector fields:

$$(3.17) \quad V_1 = \partial_x,$$

$$(3.18) \quad V_2 = \partial_t,$$

$$(3.19) \quad V_3 = x\partial_x + t\partial_t,$$

$$(3.20) \quad V_4 = x\partial_x + u\partial_u.$$

The prolongation formula up to the third order can then be listed as follows:

$$(3.21) \quad pr^{(2)}V_1 = \partial_x,$$

$$(3.22) \quad pr^{(2)}V_2 = \partial_t,$$

$$(3.23) \quad pr^{(2)}V_3 = x\partial_x + t\partial_t - u_x\partial_{u_x} - u_t\partial_{u_t} - 2u_{xx}\partial_{u_{xx}} - 2u_{xt}\partial_{u_{xt}} - 2u_{tt}\partial_{u_{tt}},$$

$$(3.24) \quad pr^{(2)}V_4 = x\partial_x + u\partial_u + u_t\partial_{u_t} - u_{xx}\partial_{u_{xx}} + u_{tt}\partial_{u_{tt}},$$

$$(3.25) \quad pr^{(3)}V_3 = pr^{(2)}V_3 - 3u_{xxx}\partial_{u_{xxx}},$$

$$(3.26) \quad pr^{(3)}V_4 = x\partial_x + u\partial_u + u_t\partial_{u_t} - u_{xx}\partial_{u_{xx}} + u_{tt}\partial_{u_{tt}}.$$

4. The group classification of the Lax–Wendroff scheme.

4.1. The Lax–Wendroff scheme. In this paper a rigorous systematic approach for the group classification of the Lax–Wendroff scheme is developed. To achieve this the difference approximations to the scalar conservation law

$$(4.1) \quad u_t + f(u)_x = 0$$

require study. The equation in (4.1) is the simplest one-dimensional nonlinear model exhibiting behavior typical of the equations of inviscid fluid dynamics. The Lax–Wendroff difference approximation to (4.1) is defined by

$$(4.2) \quad \frac{u_j^{n+1} - u_j^n}{\Delta t} + \frac{f(u_{j+1}^n) - f(u_{j-1}^n)}{2\Delta x} = \frac{1}{2}\Delta t \frac{a_{j+\frac{1}{2}}^2(u_{j+1}^n - u_j^n) - a_{j-\frac{1}{2}}^2(u_j^n - u_{j-1}^n)}{\Delta x^2},$$

where

$$\begin{aligned} u_j^n &= u(n\Delta t, j\Delta x), \\ a(u) &= \frac{\partial f(u)}{\partial u}, \\ a_{j+\frac{1}{2}} &= a\left(u_{j+\frac{1}{2}}^n\right), \\ u_{j+\frac{1}{2}}^n &= \frac{1}{2}(u_{j+1}^n + u_j^n). \end{aligned}$$

It is well known that the Lax–Wendroff scheme was designed to have the following desirable computational features [31, 32]: conservation of form; to have a three-point scheme; second order accuracy on smooth solutions. Numerical spikes and downstream oscillations are generated in the vicinity of the shock.

4.2. The modified differential equation of Lax–Wendroff scheme. Majda and Osher [32, eq. 1.5] have produced the modification equation:

$$(4.3) \quad u_t = -(f(u))_x + \frac{h^2}{6} [-(f(u))_{xxx} + \lambda^2([a'(u)((f(u))_x)^2]_x + [a(u)[a(u)(f(u))_x]_x)].$$

Some remarks are essential here:

1. The modified equation retains the conservation law form.

2. Even though the modified equation has been derived by assuming that the exact solution of (4.1) is smooth, numerical evidence indicates that, for shock solutions of (4.1), solution of the difference equation in (4.2) more closely mimics the solution of a modified equation in (4.3) than solutions of the original equation in (4.1).

Setting $f(u) = \frac{1}{2}u^2$, and performing some mechanical calculations, we arrive at the model differential equation for the Lax–Wendroff scheme [32, eq. 1.7]

$$(4.4) \quad \Delta_1 = u_t + uu_x + RHS = 0,$$

where

$$RHS = -h^2 \left[\frac{u}{6} (\lambda^2 u^2 - 1) u_{xxx} + \lambda^2 u u_x^3 + \frac{1}{4} (3\lambda^2 u^2 - 1) (2u_x u_{xx}) \right],$$

$$h = \Delta x, \quad \lambda = \frac{\Delta t}{\Delta x}.$$

4.3. Group invariants of the Lax–Wendroff scheme. Before beginning the group analysis, some detailed but mechanical calculations must be performed:

$$(4.5) \quad d_1 = \partial_x \circ \Delta_1 = 0,$$

$$(4.6) \quad d_2 = \partial_t \circ \Delta_1 = 0,$$

$$(4.7) \quad d_3 = \partial_u \circ \Delta_1 = u_x - h^2 \left[\frac{1}{6} (3\lambda^2 u^2 - 1) u_{xxx} + \lambda^2 u_x^3 + 3\lambda^2 u u_x u_{xx} \right],$$

$$(4.8) \quad d_4 = \partial_{u_x} \circ \Delta_1 = u - h^2 \left[3\lambda^2 u u_x^2 + \frac{1}{2} (3\lambda^2 u^2 - 1) u_{xx} \right],$$

$$(4.9) \quad d_5 = \partial_{u_t} \circ \Delta_1 = 1,$$

$$(4.10) \quad d_6 = \partial_{u_{xx}} \circ \Delta_1 = -h^2 \left[\frac{1}{2} (3\lambda^2 u^2 - 1) u_x \right],$$

$$(4.11) \quad d_7 = \partial_{u_{xt}} \circ \Delta_1 = 0,$$

$$(4.12) \quad d_8 = \partial_{u_{tt}} \circ \Delta_1 = 0,$$

$$(4.13) \quad d_9 = \partial_{u_{xxx}} \circ \Delta_1 = -h^2 \left[\frac{1}{6} (\lambda^2 u^3 - u) \right].$$

According to Theorem 2, the system of equations admits a space of operators with the basis V_3 (3.19) if and only if the following equations are satisfied:

$$(4.14) \quad \Delta_1 = 0,$$

$$(4.15) \quad pr^{(3)} V_3 \circ \Delta_1 = 0,$$

From (4.15), we have

$$(4.16) \quad -u_x d_4 - u_t d_5 - 2u_{xx} d_6 - 3u_{xxx} d_9 = 0.$$

Expressing the functions $d_l, l = 1, \dots, 9$, leads to a system of equations equivalent to:

$$(4.17) \quad \begin{aligned} & -u_x \left\{ u - h^2 \left[3\lambda^2 u u_x^2 + \frac{1}{2} (3\lambda^2 u^2 - 1) u_{xx} \right] \right\} - u_t \\ & - 2u_{xx} \left\{ -h^2 \left[\frac{1}{2} (3\lambda^2 u^2 - 1) u_x \right] \right\} \\ & - 3u_{xxx} \left\{ -h^2 \left[\frac{1}{6} (\lambda^2 u^3 - u) \right] \right\} = 0. \end{aligned}$$

Performing some mechanical calculations, we arrive at

$$(4.18) \quad u_{xxx} = \frac{3}{u} \frac{u_x}{(1 - \lambda^2 u^2)} [(3\lambda^2 u^2 - 1)u_{xx} + 2\lambda^2 u u_x^2].$$

The right-hand side of the group invariant equation is written entirely in terms of x derivatives. Under the condition that λ is a small parameter, (4.18) can be recast into

$$(4.19) \quad u_{xxx} \approx -\frac{3}{u} u_x u_{xx}.$$

From the theory of gas dynamics we know that, for the Navier–Stokes shock, there are some inherent constraints among the spatial derivatives: $u_x, u_{xx}, u_{xxx}, \dots$. In the following tables, the theoretical and group invariant conditions are compared, revealing a conflict.

Analysis at the upstream from the shock:

	u	u_x	u_{xx}	u_{xxx}
Theoretical solution	> 0	< 0	< 0	> 0
Lax–Wendroff scheme	> 0	< 0	< 0	< 0

Obviously, there is a conflict upstream from the shock. The signature of the third spatial derivative from group theory is inconsistent with the theoretical results, showing that this is the result of a lack of invariance in the similarity transformation upstream from the shock.

Analysis downstream from the shock:

	u	u_x	u_{xx}	u_{xxx}
Theoretical solution	> 0	< 0	> 0	> 0
Lax–Wendroff scheme	> 0	< 0	> 0	> 0

There is no conflict downstream from the shock, and the signature of the third spatial derivative from group theory is consistent with the theoretical results. The invariance of the similarity transformation is preserved upstream from the shock.

These group analysis results are consistent with previous numerical results produced by the Lax–Wendroff scheme [31, 33]. They exhibit some level of momentum spikes and postshock oscillations, while the group invariant conditions are broken in this region.

By using the same analysis, we finally obtain an exact group invariant equation for V_4

$$(4.20) \quad u_{xxx} = -\frac{3}{u} u_x u_{xx}.$$

Other results are the same as those for group analysis on the basis of V_3 . Some remarks should be made here:

1. Based on the analysis above, we can see that the origin of the spurious wave oscillations behind the shock has some inherent connection with the lack of symmetry. Obviously, there is a conflict upstream from the shock as the signature of the third spatial derivative from group theory is not consistent with the theoretical results. This is a result of the lack of invariance of the similarity transformation upstream from the shock. The Lax–Wendroff scheme exhibits some level of momentum spikes and postshock oscillations, but the group invariant conditions are broken in this region.
2. In the above analysis (4.19), it is assumed that λ is a small parameter. This condition is met in many cases, but this is not generally applicable. It appears that, in general, modification equations do not admit similarity transformations for both cases.

5. Symmetry and Harten's entropy enforcement condition. The approach to development of the shock-capturing methods TVD and essentially nonoscillatory shock-capturing method (ENO) has been primarily mathematical. Most of these schemes do not directly consider the physical reasons for oscillation production in the numerical solutions. It is well known that Harten [2] has proved that some conditions must be met for the coefficient of numerical viscosity, in order for the TVD property to hold, but the general formula of these coefficients cannot be obtained from Harten's analysis. In his celebrated paper [2], he simply presented a heuristic analysis and provided the entropy enforcement formula. In this section, we will address the coefficient of numerical viscosity from the point of view of symmetry.

5.1. TVD scheme and its modified equation. This section studies the difference approximations to the scalar conservation law and discusses numerical approximations to weak solutions of (4.1), which are obtained by $(2k+1)$ -point explicit schemes in conservation form:

$$(5.1) \quad v_j^{n+1} = v_j^n - \lambda \left(\bar{f}_{j+\frac{1}{2}}^n - \bar{f}_{j-\frac{1}{2}}^n \right),$$

where

$$\bar{f}_{j+\frac{1}{2}}^n = f(v_{j-k+1}^n, \dots, v_{j+k}^n).$$

Here $v_j^n = v(j\Delta x, n\Delta t)$, and $\bar{f}$ is a numerical flux function. The numerical flux function must be consistent with the flux $f(u)$ in the following sense:

$$(5.2) \quad \bar{f}(u, \dots, u) = f(u).$$

Let us consider a general three-point finite difference scheme in conservation form with a numerical flux $\bar{f}$ of the form [2, eq. 3.1a]

$$(5.3) \quad \bar{f}(v_j, v_{j+1}) = \frac{1}{2} \left[f(v_j) + f(v_{j+1}) - \frac{1}{\lambda} Q \left(\lambda \bar{a}_{j+\frac{1}{2}} \right) \Delta_{j+\frac{1}{2}} v \right],$$

where

$$\bar{a}_{j+\frac{1}{2}} = \frac{f(v_{j+1}) - f(v_j)}{\Delta_{j+\frac{1}{2}} v} \text{ when } \Delta_{j+\frac{1}{2}} v \neq 0,$$

TABLE 5.1

The coefficients of numerical viscosity for different conservation schemes.

Scheme	$Q(x)$
Lax-Friedrichs	$Q(x) = \frac{1}{\lambda}$
Lax-Wendroff	$Q(x) = x^2$
Roe	$Q(x) = x $
Harten	$Q(x) = \begin{cases} x , & x \geq \epsilon \\ \frac{x^2 + \epsilon^2}{2\epsilon}, & x \leq \epsilon \end{cases}$
Boris-Book	$Q(x) = x^2 + \frac{1}{4}$

$$\bar{a}_{j+\frac{1}{2}} = a(v_j) \text{ when } \Delta_{j+\frac{1}{2}} v = 0.$$

Here $Q(x)$ is a function, which is often referred to as the coefficient of numerical viscosity. The modified equation of scheme (5.3), i.e., the equation that it approximates to second order accuracy, is

$$(5.4) \quad u_t + f(u)_x = \frac{\Delta t}{2\lambda^2} \{ [Q(\nu) - \nu^2] u_x \}_x,$$

where

$$\lambda = \frac{\Delta t}{\Delta x},$$

$$\nu = \lambda a(u).$$

In Table 5.1, some typical coefficients of numerical viscosity are presented, including Harten's formula.

5.2. Group invariants of the TVD scheme. In this section, the modified equation is used instead of the first differential approximation, and the conditions under which a modified equation admits all groups of transformations allowed by the inviscid Burgers equation are formulated.

Setting $f(u) = \frac{1}{2}u^2$, we rewrite the modified equation as

$$(5.5) \quad u_t + uu_x = [D(u)u_x]_x,$$

where

$$D(u) = \frac{\Delta t}{2\lambda^2} [Q(\nu) - \nu^2].$$

Hence,

$$(5.6) \quad \Delta_2 = u_t + uu_x - D'(u)u_x^2 - D(u)u_{xx} = 0.$$

Using the same mathematical manipulation presented in section 4, we can obtain two different group invariant conditions as follows:

$$(5.7) \quad D'(u)u_x^2 + D(u)u_{xx} = 0,$$

$$(5.8) \quad [D'(u) - uD''(u)]u_x^2 + [2D(u) - uD'(u)]u_{xx} = 0.$$

Equations (5.7) and (5.8) consist of a homogeneous equation system about variables u_x^2, u_{xx} . The nondegenerate conditions for these variables are

$$(5.9) \quad \begin{vmatrix} D'(u) & D(u) \\ D'(u) - uD''(u) & 2D(u) - uD'(u) \end{vmatrix} \\ = uD(u)D''(u) + D(u)D'(u) - uD'^2(u) = 0.$$

This is a nonlinear equation of variable u ; the function is $D(u)$. The solution of this equation is

$$(5.10) \quad D(u) = D_0 u^a,$$

where D_0, a are constant parameters.

According to Harten's definition [2, eq. 3.15b], we have

$$(5.11) \quad Q(\nu) = \nu^2 + \frac{2\lambda^2}{\Delta t} D(u).$$

For the present problem,

$$(5.12) \quad \nu = \lambda u.$$

Using the group invariant condition (5.10), we have

$$(5.13) \quad Q(\nu) = \nu^2 + \frac{2\lambda^2}{\Delta t} D_0 u^a$$

or

$$(5.13') \quad Q(\nu) = \nu^2 + \frac{2\lambda^2}{\Delta t} D_0 \lambda^{-a} \nu^a.$$

In order to obtain a universal formula the coefficients in it must be dimensionalless parameters. Δx and Δt denote, as usual, the dimensions of length and time, respectively. It is clear that D_0 can be rewritten as

$$(5.14) \quad D_0 = \hat{\varepsilon}(\Delta x)^b(\Delta t)^c,$$

where $\hat{\varepsilon}$ is a dimensionless parameter; meanwhile b, c are the exponents that must be chosen.

Finally, we have

$$(5.15) \quad Q(\nu) = \nu^2 + 2\hat{\varepsilon}\nu^a.$$

These dimensional exponents must be met:

$$(5.16) \quad \begin{aligned} Q_0 &= (\Delta x)^b(\Delta t)^c(\Delta t)^{-1}(\Delta t)^{2-a}(\Delta x)^{a-2} \\ &= (\Delta x)^{a+b-2}(\Delta t)^{c-a+1}. \end{aligned}$$

Hence:

$$(5.17) \quad a + b - 2 = 0,$$

$$(5.18) \quad c - a + 1 = 0.$$

That is,

$$(5.19) \quad b = 2 - a,$$

$$(5.20) \quad c = a - 1.$$

These exponents form the universal formula of the coefficients of numerical viscosity, inheriting the symmetries of the PDE. Different schemes could be deduced depending upon the choice of the parameter a .

Another dissipation condition reads as:

$$(5.21) \quad Q(0) > 0.$$

So, the choice of the parameter a is

$$(5.22) \quad a = 0.$$

Finally, we have a definition of the coefficients of numerical viscosity, deduced from the symmetries:

$$(5.23) \quad Q(x) = x^2 + 2\hat{\varepsilon}.$$

THEOREM 3. *If we let $Q(x)$ in (5.3) satisfy the equality*

$$Q(x) = x^2 + 2\hat{\varepsilon},$$

where $\hat{\varepsilon} > 0$ is a dimensionless constant, then the finite different scheme

$$\hat{f}_{j+\frac{1}{2}} = \frac{1}{2} \left[f(u_j^n) + f(u_{j+1}^n) - \frac{1}{\lambda} Q(\lambda \bar{a}_{j+\frac{1}{2}}^n) (u_{j+1}^n - u_j^n) \right]$$

satisfies all of the group invariant conditions of the nonlinear wave equation.

6. The reexploration of the spatial oscillations in finite difference solutions for Navier–Stokes shocks. In the calculation of complex flow fields containing shock waves, much attention has been paid to the shock capturing method [1, 2, 3, 4, 5, 6, 7, 8, 9]. In order to capture shock waves smoothly without spurious data near or in the shock region, the development of high-resolution nonoscillatory dissipative schemes containing no free parameters has been much emphasized. These include TVD schemes and ENO schemes. In [4, 36], Zhang found that, with a linearized analysis of one-dimensional Navier–Stokes equations and a proper manipulation of the coefficients of third order derivative ν_3 , the spurious oscillation both upstream and downstream from shocks can be totally suppressed. The choice of ν_3 must be in accordance with the second law of thermodynamics. The numerical experiments with nonlinear Navier–Stokes equations confirm that conclusions reached through linearized analysis are valid in general. The importance of the role played by ν_3 has not been previously recognized [35]. While it has been usually thought that the third order dispersion terms in modified differential equations would not play an essential role in a linear problem, e.g., in stability analysis, etc., they do contribute through nonlinear interactions to the unwanted wave structure and even lead to chaos, if they are not properly treated. Based on the above physical concepts, Zhang suggested the celebrated NND schemes that have been applied to the numerical simulation of Navier–Stokes equations with some success [4, 35].

It is well known that, in application, Navier–Stokes equations are unsteady and nonlinear. The validity of the above linearized analysis is an open question. Difficulty in nonlinear analysis is primarily caused by the nonlinearity inherent in the Navier–Stokes equations, in particular the mechanism that allows the formation of shocks. The theme of this section is to emphasize the role of symmetry in various aspects of a proper manipulation of the coefficients of third order derivative ν_3 .

6.1. The model equation and its symmetries. To avoid complicated mathematics, let us consider only the one-dimensional Navier–Stokes shock equation [36]:

$$(6.1) \quad \rho u_t + \frac{\gamma+1}{2\gamma} \rho_\infty u_\infty \left(1 - \frac{u_2 u_\infty}{u^2}\right) u_x = \frac{4}{3} (\mu u_x)_x,$$

where ρ is the mass per unit volume, u is the velocity, t is time, x is the spatial coordinate, γ is the specific heat ratio, and μ is dynamic viscosity.

Here

$$\begin{aligned} \rho &= \frac{\rho_\infty u_\infty}{u}, \\ \frac{\mu}{\mu_\infty} &= \left(\frac{h}{h_\infty}\right)^n, \\ h &= \frac{1}{2} u_\infty^2 \left(1 + \frac{2}{\gamma-1} \frac{1}{M_\infty^2}\right) - \frac{1}{2} u^2. \end{aligned}$$

For the propagating shock problem the Dirichlet boundary conditions for u are required up- and downstream from the shock. These are

$$(6.2) \quad \begin{aligned} u &= u_\infty, & x &\rightarrow -\infty, \\ u_2 &= \frac{\gamma-1}{\gamma+1} \left(1 + \frac{2}{\gamma-1} \frac{1}{M_\infty^2}\right) u_\infty. \end{aligned}$$

The system of equations (6.1) admits a space of operators with the basis:

1. translation in time $V_1 = \partial_t$,
2. translation in a space coordinate $V_2 = \partial_x$,
3. Galileo transformation $V_3 = t\partial_x + \partial_u$,
4. similarity transformation $V_4 = x\partial_x + t\partial_t$,

together with the prolongation vectors as follows, only showing the spatial partial derivatives of u :

$$(6.3) \quad pr^{(3)}V_1 = \partial_t,$$

$$(6.4) \quad pr^{(3)}V_2 = \partial_x,$$

$$(6.5) \quad pr^{(3)}V_3 = t\partial_x + \partial_u - u_x \partial_{u_t},$$

$$(6.6) \quad pr^{(3)}V_4 = x\partial_x + t\partial_t - u_x \partial_{u_x} - u_t \partial_{u_t} - 2u_{xx} \partial_{u_{xx}} - 3u_{xxx} \partial_{u_{xxx}}.$$

6.2. The modified equation and its symmetries. It can be stated that, generally when the leading term of the truncation error is the third order derivative, the modified equation of the above one-dimensional Navier–Stokes shock is

$$(6.7) \quad \rho u_t + \frac{\gamma+1}{2\gamma} \rho_\infty u_\infty \left(1 - \frac{u_2 u_\infty}{u^2}\right) u_x = \frac{4}{3} (\mu u_x)_x - (\nu_3 u_{xx})_x.$$

The physical viscous coefficient will be taken as constant wherever convenient, and the modified equation could be written as follows:

$$(6.8) \quad \Delta_3 = u_t + \frac{\gamma+1}{2\gamma} \left(u - \frac{u_2 u_\infty}{u} \right) u_x - \frac{\nu}{\rho_\infty u_\infty} u u_{xx} + \frac{\nu_3}{\rho_\infty u_\infty} u u_{xxx} = 0.$$

By using the same mathematical manipulation presented in section 4, we can obtain two different group invariance conditions listed as follows:

(1) For the invariants of the Galileo transformation, the group invariance condition reads

$$(6.9) \quad \nu_3 u_{xxx} = \nu u_{xx} + \rho_\infty u_\infty \left(\frac{\gamma-1}{2\gamma} - \frac{\gamma+1}{2\gamma} \frac{u_2 u_\infty}{u^2} \right) u_x.$$

(2) For the invariants of the similarity transformation, the group invariance condition reads:

$$(6.10) \quad \nu_3 u_{xxx} = \frac{1}{2} \nu u_{xx}.$$

6.3. The choice of ν_3 in accordance with the group invariants. First, considering the invariants of the similarity transformation, the corresponding signature of ν_3 can be calculated as follows.

Equation (6.10) can be recast into

$$(6.11) \quad \nu_3 = \frac{\nu}{2} \cdot \frac{u_{xx}}{u_{xxx}}.$$

Hence, we have

$$(6.12) \quad \text{signature}[\nu_3] = \text{signature}[u_{xx}] \times \text{signature}[u_{xxx}].$$

From (6.12), we have

$$(6.13) \quad \nu_3 < 0$$

in the upstream region, while

$$(6.14) \quad \nu_3 > 0$$

in the downstream region.

ν_3 is chosen using the similarity invariance analysis developed above. Now it is obvious that, if we can keep $\nu_3 < 0$ upstream and $\nu_3 > 0$ downstream, the invariants of the similarity transformation condition can be met. This result is equivalent to the former linearized result.

6.4. The Galileo invariants. The invariants of the Galileo transformation are more complex than those of the similarity transformations. In general conditions, there is not a simple relation for the choice of ν_3 . In this situation, it is thought that the invariants of the similarity transformations are met, and those for the Galileo transformation are not. Here we focus only on the region near the shock. In this region, we have

$$(6.15) \quad \frac{\gamma-1}{2\gamma} - \frac{\gamma+1}{2\gamma} \frac{u_2 u_\infty}{u^2} \approx \frac{\gamma-1}{2\gamma} - \frac{\gamma+1}{2\gamma} \frac{u_2}{u_\infty}.$$

The speed of shock propagation is the governing parameter for this problem. For a given speed of shock propagation, the boundary value of u_2 is obtained from the Rankine–Hugoniot relation:

$$(6.16) \quad \frac{u_2}{u_\infty} = \frac{\gamma - 1}{\gamma + 1} \left(1 + \frac{2}{\gamma - 1} \frac{1}{M_\infty^2} \right).$$

Hence,

$$(6.17) \quad \begin{aligned} \frac{\gamma - 1}{2\gamma} - \frac{\gamma + 1}{2\gamma} \frac{u_2 u_\infty}{u^2} &\approx \frac{\gamma - 1}{2\gamma} - \frac{\gamma + 1}{2\gamma} \frac{u_2}{u_\infty} \\ &= \frac{\gamma - 1}{2\gamma} - \frac{\gamma + 1}{2\gamma} \frac{\gamma - 1}{\gamma + 1} \left(1 + \frac{2}{\gamma - 1} \frac{1}{M_\infty^2} \right) \\ &= \frac{\gamma - 1}{2\gamma} - \frac{\gamma - 1}{2\gamma} - \frac{1}{\gamma M_\infty^2} \\ &= -\frac{1}{\gamma M_\infty^2}. \end{aligned}$$

The Galileo invariants can be written in compact form as:

$$(6.18) \quad \nu_3 u_{xxx} = \nu u_{xx} - \frac{1}{\gamma M_\infty^2} \rho_\infty u_\infty u_x.$$

If $M_\infty \gg 1$, then the choice of ν_3 is the same as that of the similarity transformation. But, for the case $M_\infty \ll 1$,

$$(6.19) \quad \nu_3 u_{xxx} = -\frac{1}{\gamma M_\infty^2} \rho_\infty u_\infty u_x.$$

The Galileo invariants are not met up- or downstream.

From the above consideration, we know that for fast shock propagation the choice of ν_3 based on the linearized analysis is valid under unsteady, nonlinear conditions, but for slow shock propagation, the Galileo invariants are not met in some cases.

It should be remarked that:

A NND scheme has been suggested based on linear analysis, which has good stability characteristics and convergence. Through the preliminary numerical calculations, confidence was gained in the use of the NND schemes. The principle by which the coefficients of the third order derivatives are chosen is still based on the linearized conclusion. Under unsteady, nonlinear conditions, the validity of this linear conclusion is an open question. The use of symmetries helps us to reach a new conclusion and demonstrate that the linear conclusion is met with the similarity invariants but not met with the Galileo invariants in the general case.

7. Conclusions. It is known that the numerical solutions calculated by finite difference schemes are always associated with numerical dissipation and dispersion. Such errors can lead to undesirable numerical effects, especially for shock capturing. This paper uses a Lie symmetry analysis method to investigate some noteworthy properties of several difference schemes for nonlinear equations. We have demonstrated that the loss of properties of invariance from a modification equation, which conserve the main properties of its difference scheme, can serve as the cause of nonphysical

oscillations that are capable of eventually completely destroying a numerical solution. Further, this points to symmetry breaking as the inherent cause of nonphysical oscillation in the vicinity of shock. Each of these results is vital to the further development of the application of Lie symmetry to the study of shock-capturing methods.

It should be pointed out that the conclusions presented in this paper have preliminary character and demand further study.

Acknowledgments. I appreciate the numerous corrections and very useful comments of the referees. Special thanks to the referee who provided valuable information about the history of this research.

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